

2. Part B: Managing Complications during Pregnancy, Childbirth and Postnatal period

2.1 Haemorrhage

2.1.1 WHO recommendations for the prevention and treatment of postpartum haemorrhage

(44)

New

Strong Recommendation, moderate low-quality evidence

Intravenous oxytocin is the recommended uterotonic drug for the treatment of PPH.

Strong Recommendation, low-quality evidence

If intravenous oxytocin is unavailable, or if the bleeding does not respond to oxytocin, the use of intravenous ergometrine, oxytocin-ergometrine fixed dose, or a prostaglandin drug (including sublingual misoprostol, 800 µg) is recommended.

- The GDG recommended IV oxytocin as the first line uterotonic drug for the treatment of PPH, including when women have already received this drug for the prophylaxis of PPH.
- The GDG recognized that IV oxytocin may not be available in all settings. It encourages health care decision-makers in these settings to strive to make oxytocin available.
- In settings where IV oxytocin is unavailable to women who have received prophylactic IM oxytocin during the third stage of labour, the GDG considered misoprostol to be a valid alternative.
- If PPH prophylaxis with misoprostol has been administered and if injectable uterotonics are unavailable, there is insufficient evidence to guide further misoprostol dosing and consideration must be given to the risk of potential toxicity.
- There is no added benefit to offering misoprostol simultaneously to women receiving oxytocin for the treatment of PPH (i.e. adjunct misoprostol).
- The GDG noted that the two largest trials of misoprostol for the treatment of PPH (Winikoff 2010, Blum 2010) reported the use of a 800 µg dose administered sublingually. The majority of the GDG members agreed that 800 µg is an acceptable sublingual misoprostol dose for the treatment of PPH, though some members of the GDG expressed concern related to the risk of hyperpyrexia associated with this dosage.
- If IV oxytocin has been used for the treatment of PPH and the bleeding does not stop, there is a paucity of data to recommend preferences for second line uterotonic drug treatment. Decisions in such situations must be guided by the experience of the provider, the availability of the drugs, and by known contraindications.
- In situations in which IM oxytocin can be administered and there is no possibility of IV treatment with ergot alkaloids/ injectable prostaglandins, there is a paucity of data to recommend a preference of IM oxytocin over misoprostol or other uterotonics. Decisions in such situations must be guided by the experience of the provider, the availability of the drugs, and by known contraindications.

New

Strong Recommendation, low-quality evidence

The use of isotonic crystalloids is recommended in preference to the use of colloids for the intravenous fluid resuscitation of women with PPH.

Weak Recommendation, moderate-quality evidence

The use of tranexamic acid is recommended for the treatment of PPH if oxytocin and other uterotonics fail to stop the bleeding or if it is thought that the bleeding may be partly due to trauma.

- Evidence for the recommendation of tranexamic acid was extrapolated from the literature on surgery and trauma, which shows tranexamic acid to be a safe option for the treatment of trauma-related bleeding.

Strong Recommendation, very low-quality evidence

Uterine massage is recommended for the treatment of PPH.

Weak Recommendation, very low-quality evidence

If women do not respond to treatment using uterotonics, or if uterotonics are unavailable, the use of intrauterine balloon tamponade is recommended for the treatment of PPH due to uterine atony.

Weak Recommendation, very low-quality evidence

If other measures have failed and if the necessary resources are available, the use of uterine artery embolization is recommended as a treatment for PPH due to uterine atony.

Strong Recommendation, very low-quality evidence

If bleeding does not stop in spite of treatment using uterotonics and other available conservative interventions (e.g. uterine massage, balloon tamponade), the use of surgical interventions is recommended.

Weak Recommendation, very low-quality evidence

The use of bimanual uterine compression is recommended as a temporizing measure until appropriate care is available for the treatment of PPH due to uterine atony after vaginal delivery.

Weak Recommendation, very low-quality evidence

The use of external aortic compression for the treatment of PPH due to uterine atony after vaginal birth is recommended as a temporizing measure until appropriate care is available.

Weak Recommendation, very low-quality evidence

The use of non-pneumatic anti-shock garments is recommended as a temporizing measure until appropriate care is available.

Weak Recommendation against, very low-quality evidence

The use of uterine packing is not recommended for the treatment of PPH due to uterine atony after vaginal birth.

- The GDG noted that the application of these interventions requires training and that maternal discomfort and complications associated with these procedures have been reported.
- Uterine massage as a therapeutic measure is defined as the rubbing of the uterus achieved through the manual massaging of the abdomen. This is typically sustained until the bleeding stops or the uterus contracts. The GDG considered that uterine massage should be started once PPH has been diagnosed.
- The initial rubbing of the uterus and expression of blood clots are not regarded as therapeutic uterine massage.
- When rating the recommendation #17 as 'strong', the low cost and safety of uterine massage were taken into account.
- The use of balloon tamponade was considered by the GDG to be a measure that can potentially avoid surgery or as a temporizing measure while awaiting transfer to a higher level facility. The GDG acknowledges that balloon tamponade can be obtained with specific devices as well as with lower cost adaptations, including those based on the use of condoms and surgical gloves.
- The GDG noted that uterine artery embolization requires significant resources, in terms of the cost of the treatment, the facilities, and the training of health care workers.
- The GDG noted that conservative surgical approaches should be tried first. If these do not work, they should be followed by more invasive procedures. Compression sutures, for example, may be attempted as a first intervention, and if these fail, then uterine, utero-ovarian and hypogastric vessel ligation may be tried. If life-threatening bleeding continues even after ligation, then a subtotal (otherwise known as supracervical) or total hysterectomy should be performed.
- The GDG acknowledged that the level of health care provider skills will play a role in the selection and sequence of the surgical interventions.
- External aortic compression has long been recommended as a potential life-saving technique, and mechanical compression of the aorta, if successful, slows blood loss. The GDG placed a high value on this procedure as a temporizing measure in the treatment of PPH.
- The GDG noted that research evaluating the potential benefits and harms of non-pneumatic anti-shock garments is ongoing. Based on the evidence available, the GDG regarded non-pneumatic anti-shock garments as a temporizing measure while transfer is awaited.
- The GDG noted that there was no evidence of benefit of uterine packing and placed a high value on concerns

regarding its potential harm.

New

Weak Recommendation, very low-quality evidence

If the placenta is not expelled spontaneously, the use of additional oxytocin (10 IU, IV/IM) in combination with controlled cord traction is recommended.

Weak Recommendation against, very low-quality evidence

The use of ergometrine for the management of a retained placenta is not recommended as this may cause tetanic uterine contractions which may delay the expulsion of the placenta.

Weak Recommendation against, very low-quality evidence

The use of prostaglandin E2 alpha (dinoprostone or sulprostone) in the management of retained placenta is not recommended.

Weak Recommendation, very low-quality evidence

A single dose of antibiotics (ampicillin or first-generation cephalosporin) is recommended if manual removal of the placenta is practised.

- The GDG found no empirical evidence to support recommending the use of uterotonics for the management of a retained placenta in the absence of haemorrhage. The above recommendation was reached by consensus.
- The WHO guide, "Managing complications in pregnancy and childbirth" (52), states that if a placenta is not expelled within 30 minutes after the delivery of a baby, the woman should be diagnosed as having a retained placenta. Since there is no evidence for or against this definition, the delay used before this condition is diagnosed is left to the judgement of the clinician.
- The same WHO guide also suggests that in the absence of haemorrhage, the woman should be observed for a further 30 minutes after the initial 30 minutes, before the manual removal of the placenta is attempted. The GDG noted that spontaneous expulsion of the placenta can still occur, even in the absence of bleeding. A conservative approach is therefore advised and the timing of the manual removal of the placenta as a definitive treatment is left to the judgement of the clinician.
- The recommendation regarding the use of prostaglandin E2 is informed by a lack of evidence on this question and also by concerns related to adverse events, particularly cardiac events.
- Direct evidence of the value of antibiotic prophylaxis after the manual removal of the placenta was not available. The GDG considered indirect evidence of the benefit of prophylactic antibiotics from studies of caesarean section and abortion, as well as observational studies of other intrauterine manipulations.
- Current practice suggests that ampicillin or first-generation cephalosporins may be administered when the manual removal of the placenta is performed.
- This question was identified as a research priority for settings in which prophylactic antibiotics are not routinely administered and those with low infectious morbidity.

New

Strong Recommendation, moderate-quality evidence

In settings where skilled birth attendants are not present and oxytocin is unavailable, the administration of misoprostol (600 µg PO) by community health care workers and lay health workers is recommended for the prevention of PPH.

Weak Recommendation, high-quality evidence

In settings where skilled birth attendants are available, controlled cord traction (CCT) is recommended for vaginal births if the care provider and the parturient woman regard a small reduction in blood loss and a small reduction in the duration of the third stage of labour as important.

Strong Recommendation against, moderate-quality evidence

In settings where skilled birth attendants are unavailable, CCT is not recommended.

New

Strong Recommendation, moderate-quality evidence

Late cord clamping (performed after 1 to 3 minutes after birth) is recommended for all births while initiating simultaneous essential newborn care.

Strong Recommendation against, moderate-quality evidence

Early cord clamping (<1 minute after birth) is not recommended unless the neonate is asphyxiated and needs to be moved immediately for resuscitation.

New

Strong Recommendation, very low-quality evidence

Postpartum abdominal uterine tonus assessment for early identification of uterine atony is recommended for all women.

New

Strong Recommendation, moderate-quality evidence

CCT is the recommended method for removal of the placenta in caesarean section.

2.1.2 WHO recommendation on umbilical vein injection of oxytocin for the treatment of retained placenta

(281)

Research-context recommendation

Umbilical vein injection of oxytocin is recommended for the treatment of retained placenta only in the context of rigorous research.

Remarks

- *The Guideline Development Group acknowledged the potential of umbilical vein injection of oxytocin in the treatment of retained placenta but considered the evidence of benefit in terms of manual removal of the placenta without impact on other priority outcomes insufficient to make a recommendation for routine clinical practice. The group agreed that high-quality randomized trials comparing umbilical vein injection of uterotonics with expectant management of women with retained placenta are needed, with the aim of demonstrating its impact on severe morbidity related to postpartum haemorrhage in addition to a reduction in manual removal of the placenta.*
- *When used in a research context, it is safer to consider the use of this intervention in situations in which retained placenta occurs in the absence of abnormal bleeding.*
- *There are three types of retained placenta, and umbilical vein injection is likely to be only effective in placenta adherens, the most common type of retained placenta, which occurs as a result of failed contraction of the retroplacental myometrium. To date, studies have not distinguished the subtypes before treatment, and this may have contributed to the results showing lack of efficacy of treatment with umbilical vein injection for retained placenta.*

Justification

Evidence from trials that compared both umbilical vein injection of oxytocin versus expectant management and umbilical vein injection of oxytocin versus umbilical vein injection of saline suggest that this intervention may lead to a reduction in manual removal of placenta. However, the effect of this intervention on other priority outcomes (including infections, maternal satisfaction and length of hospitalization) is unclear. While the cost-effectiveness is not known, additional costs in supplies required to implement this intervention are probably negligible. When compared with injection of other solutions and uterotonics, no other umbilical vein injection regimen was shown to be clearly better than umbilical vein injection of oxytocin.

2.2 Hypertensive Disorders

2.2.1 WHO recommendations for prevention and treatment of pre-eclampsia and eclampsia

(282)

New

Strong Recommendation, moderate-quality evidence

Low-dose acetylsalicylic acid (aspirin, 75 mg/day) is recommended for the prevention of pre-eclampsia in women at high risk of developing the condition.

Weak Recommendation, low-quality evidence

Low-dose acetylsalicylic acid (aspirin, 75 mg/day) for the prevention of pre-eclampsia and its related complications should be initiated before 20 weeks of pregnancy.

- *Women are regarded as being at high risk of developing pre-eclampsia if they have one or more of the following risk factors: previous preeclampsia; diabetes; chronic hypertension; renal disease; autoimmune disease; and multiple pregnancies. This is not an exhaustive list, but can be adapted/complemented based on the local epidemiology of pre-eclampsia.*
- *The guideline development group acknowledged that in settings where 75 mg aspirin tablets are not available, the available dose nearest to 75 mg should be used.*
- *While low-dose aspirin has been shown to be beneficial in women at high risk of preeclampsia, there is a paucity of evidence to suggest that any subset of women within the high-risk group would benefit from aspirin therapy.*
- *The guideline development group noted that it may be appropriate to initiate antiplatelet agents before 20 weeks of gestation, and, if possible, as early as 12 weeks of gestation.*

New

Weak Recommendation against, low-quality evidence

Advice to rest at home is not recommended as an intervention for the primary prevention of pre-eclampsia and hypertensive disorders of pregnancy in women considered to be at risk of developing those conditions.

Weak Recommendation against, low-quality evidence

Strict bedrest is not recommended for improving pregnancy outcomes in women with hypertension (with or without proteinuria) in pregnancy.

- *The guideline development group acknowledged that there may be situations in which different levels of rest, either at home or in hospital, may be indicated for individual women. The above recommendations do not cover advice regarding overall physical activity and manual or office work.*
- *Women may need to be hospitalized for reasons other than bedrest, such as for maternal and fetal surveillance. The guideline development group agreed that hospitalization for maternal and fetal surveillance is resource intensive and should be considered as a priority for research and future recommendations.*

New

Weak Recommendation against, moderate-quality evidence

Restriction in dietary salt intake during pregnancy with the aim of preventing the development of pre-eclampsia and its complications is not recommended.

- *The guideline development group agreed that healthy dietary practices should be promoted in the general population, including among pregnant women.*
- *The group considered the avoidance of excessive dietary salt intake as a healthy dietary practice.*

Strong recommendation against, low-quality evidence

Diuretics, particularly thiazides, are not recommended for the prevention of preeclampsia and its complications.

- *The guideline development group considered that there is absence of clinical uncertainty over whether treatment of severe hypertension during pregnancy is beneficial. This recommendation was made based on expert opinion; the group considered that most maternal deaths related to hypertensive disorders are associated with complications of uncontrolled severe high blood pressure. Based on that, the group agreed that antihypertensive treatment should be recommended in all cases of severe acute hypertension.*
- *With regard to the treatment of mild/moderate hypertension in pre-eclampsia, a formal evidence review was conducted. The guideline development group considered the available evidence controversial, as there are potential harms and benefits associated with both lines of action. The group was aware of ongoing trials that might provide more robust data in the near future for guidance. Hence, they decided not to issue a recommendation on the treatment of mild/moderate hypertension until further evidence becomes available.*
- *In terms of the choice and route of administration of an antihypertensive drug for severe hypertension during pregnancy, the guideline development group noted that not only is the evidence base for this recommendation limited, but also some antihypertensive drugs may not be feasible options in many settings. The group acknowledged that hydralazine, alpha methyldopa, beta blockers (including labetalol) and nifedipine have been extensively used, and therefore, these agents would seem to be reasonable choices until further evidence becomes available. The group noted that there was no evidence to suggest that nifedipine interacts adversely with magnesium sulfate. In addition, the group considered that the use of angiotensin-converting enzyme inhibitors, angiotensin receptor blockers and sodium nitroprusside should be avoided due to safety concerns.*
- *In not recommending diuretics, particularly thiazides, for the prevention of preeclampsia and its complications, the group noted that this recommendation applies only to women at risk of developing preeclampsia who are not currently under treatment with diuretics. It does not apply to the use of diuretics for non-pre-eclampsia-related indications.*

Strong Recommendation, high-quality evidence

Magnesium sulfate is recommended for the prevention of eclampsia in women with severe pre-eclampsia in preference to other anticonvulsants.

Strong Recommendation, moderate-quality evidence

Magnesium sulfate is recommended for the treatment of women with eclampsia in preference to other anticonvulsants.

Strong Recommendation, moderate-quality evidence

The full intravenous or intramuscular magnesium sulfate regimens are recommended for the prevention and treatment of eclampsia.

Strong Recommendation, very low-quality evidence

For settings where it is not possible to administer the full magnesium sulfate regimen, the use of magnesium sulfate loading dose followed by immediate transfer to a higher level health-care facility is recommended for women with severe pre-eclampsia and eclampsia.

- *Magnesium sulfate is a lifesaving drug and should be available in all health-care facilities throughout the health system. The guideline development group believed that capacity for clinical surveillance of women and administration of calcium gluconate were essential components of the package of services for the delivery of magnesium sulfate.*
- *Clinical evidence supports the use of magnesium sulfate in all pre-eclampsia patients. In settings where there are resource constraints to manage the administration of magnesium sulfate safely in all women with pre-eclampsia, there may be a need to accord greater priority to the more severe cases. Magnesium sulfate is effective in preventing seizures in both mild and severe pre-eclampsia. However, the guideline development group noted that a higher number of women need to be treated to prevent one seizure. The group agreed on the need to treat women with severe preeclampsia, but the group members were divided on the use of magnesium sulfate as a prophylaxis for mild pre-eclampsia.*
- *Large trials have evaluated and demonstrated the effectiveness of full regimens of magnesium sulfate, which include a loading dose followed by 24-hour maintenance therapy. Specific guidance on how to administer magnesium sulfate can be found in the WHO manual entitled Managing complications in pregnancy and childbirth: a guide for midwives and doctors (283).*
- *The guideline development group deliberated on the best course of action in settings in which it is not possible to administer the full magnesium sulfate regimen. The group debated the possible (but yet unproven) benefits of administering only the loading dose versus transferring women with severe preeclampsia and eclampsia without any magnesium sulfate. The group felt that that, even in cases where immediate transfer of the woman to a higher-level facility was not possible, the patient was likely to be better off with only the loading dose than without it. The group felt that since this was a common scenario in many low-income countries, it should be given high priority for further research.*

New

Strong Recommendation, low-quality evidence

In women with severe pre-eclampsia at term, a policy of early delivery is recommended.

Weak Recommendation, moderate-quality evidence

In women with mild pre-eclampsia or gestational hypertension at term, induction of labour is recommended.

- *The guideline development group considered that there is absence of clinical uncertainty over whether termination of pregnancy in women with severe pre-eclampsia at term is beneficial. Quality of evidence provided by the Hypitad trial (284) further downgraded for indirectness.*
- *The guideline development group considered that, in women with pre-eclampsia at term, expectant management is associated with a substantial risk of further maternal and fetal complications and absence of substantial maternal and fetal benefits.*
- *In settings where gestational age is difficult to determine accurately, special attention should be paid to avoid iatrogenic prematurity in infants.*
- *The guideline development group considered that, if induction of labour is contraindicated due to maternal or fetal conditions, early delivery by caesarean section is recommended (as opposed to expectant management).*

New

Strong Recommendation, very low-quality evidence

In women treated with antihypertensive drugs antenatally, continued antihypertensive treatment postpartum is recommended.

Strong Recommendation, very low-quality evidence

Treatment with antihypertensive drugs is recommended for severe postpartum hypertension.

- *The guideline development group recognized the need for discharge instructions, including education concerning the signs and symptoms associated with postpartum hypertension.*
- *In women receiving postpartum antihypertensive treatment, at the present time it is not known at what point the treatment and monitoring of hypertension could be stopped. Hence, the group highlighted this topic as a research priority.*
- *The guideline development group put more emphasis on the frequency of postpartum deaths related to stroke and recognized that the maximum increase in blood pressure usually occurs towards the end of the first postpartum week (when, in most settings, women have been already discharged from facility care).*
- *In women diagnosed with mild pre-eclampsia antenatally, but not treated with antihypertensive drugs, the initiation of antihypertensive treatment postpartum should be considered for minimizing the risk of complications of severe high blood pressure (see remark 'c' above). That remark was made based on expert opinion and considering the evidence related to the treatment of mild/moderate hypertension during pregnancy. In the postpartum period, the maternal risk of a complication of hypertension is not counterbalanced by the risk of an adverse fetal effect produced by maternal hypotension.*
- *The guideline development group considered that there is little clinical uncertainty over whether treatment of severe postpartum hypertension is beneficial. This recommendation was made based on expert opinion and the guideline development group considered that most maternal deaths related to hypertensive disorders are associated with complications of uncontrolled severe high blood pressure. Based on that, the guideline development group agreed that antihypertensive treatment should be recommended in all cases of severe acute hypertension.*

Weak Recommendation against, very low-quality evidence

The use of corticosteroids for the specific purpose of treating women with HELLP syndrome is not recommended.

- *The guideline development group noted that, in addition to the existing evidence, three small trials addressing this research question had been registered in the WHO International Clinical Trials Registry Platform (285). In one trial (66 women) recruitment had been completed, in the second trial it was still ongoing (160 women) and in the third recruitment was yet to begin. In view of the very low quality of the evidence base on this topic and relative ease of use and availability/affordability of corticosteroids, the group accorded corticosteroids for the treatment of HELLP syndrome high priority for further research.*
- *The guideline development group emphasized that the use of corticosteroids for other indications, such as fetal lung maturation, are not included in the above recommendation.*

2.2.2 WHO recommendations on drug treatment for non-severe hypertension in pregnancy

(286)

Context specific recommendation

Women with non-severe hypertension during pregnancy should be offered antihypertensive drug treatment in the context of good quality antenatal care follow-up.

Context specific recommendation

Oral alpha-agonist (methyldopa) and beta-blockers should be considered as effective treatment options for non-severe hypertension during pregnancy.

Remarks

- *The Guideline Development Group (GDG) considered that while the use of an antihypertensive drug for the treatment of non-severe hypertension in pregnancy may confer health benefits, pregnant women who are prescribed these drugs require regular outpatient monitoring and review by an antenatal care provider. Access to antenatal care services for monitoring of blood pressure and complications (such as proteinuria), or side-effects due to treatment, is considered integral to initiating antihypertensive treatment.*
- *The GDG acknowledged that, based on available evidence, alpha-agonist (methyldopa) and beta-blockers are reasonable antihypertensive drug treatment options. The group considered it important that clinicians select an antihypertensive drug regimen appropriate to the woman's individual clinical situation. The choice of antihypertensive should be based on pre-existing antihypertensive treatment, side-effect profiles, risks (including potential fetal effects), cost, local availability and the woman's preferences. Methyldopa has the fewest safety concerns, is listed for use as an antihypertensive agent during pregnancy in the WHO Model List of Essential Medicines, and is widely available in many countries. Available evidence suggests that calcium channel blockers should be avoided.*
- *Available trials used several different oral beta-blockers (including acebutolol, atenolol, labetalol, mepindolol, metoprolol, oxprenolol, pindolol and propranolol) at different doses. It is therefore not possible to determine the optimal beta-blocker option or dosing regimen for this indication. Atenolol and metoprolol are listed on the WHO Model List of Essential Medicines and are widely available in many countries.*
- *The use of angiotensin-converting enzyme inhibitors, angiotensin receptor blockers and sodium nitroprusside should be avoided due to safety concerns.*

Justification

- When used for non-severe hypertension in pregnancy, use of an antihypertensive drug compared to placebo or no antihypertensive treatment probably reduces the development of severe hypertension, though there may be little or no difference in the risk of developing proteinuria or pre-eclampsia. There may be a slight increase in side-effects with the use of an antihypertensive drug.
- Several antihypertensive drug options have been evaluated for non-severe hypertension in pregnancy, though there is